# Tree type NN: exploring double descent and other aspects

**Aroma Indus, IIT Delhi** 22 October 2026

**Summary.** On CIFAR-10, augmentation combined with a correction tree achieves zero errors on 45,000 original training images and 74.45% test accuracy, compared with 72.37% for the clean root baseline. Five-member voting reaches 74.99%. Empirical prediction variance and a squared-bias proxy decrease with more head-training samples. On the audited Cats vs Dogs split, augmentation raises test accuracy from 80.05% to 81.97%, and five-tree voting reaches 82.85%. Neither dataset establishes interpolation-centered double descent. Local loss slices show a mismatch between training fit and validation loss.

## 1. Architecture and experimental protocol

CIFAR-10 uses 45,000 training and 5,000 validation images from a seed-42 permutation of the official training pool, plus the official 10,000-image test set. Inputs are RGB, $32 \times 32$. Two Conv–ReLU–MaxPool blocks (channels $3 \to 16 \to 32$, $3 \times 3$ kernels, padding 1) yield 2,048 features. Each class root is an MLP $2048 \to 128 \to 1$ with ReLU. Ten roots train with weighted one-versus-rest binary cross-entropy (BCE); prediction is the largest raw score.

**Augmentation and caching.** Both conditions start from the same earlier CNN checkpoint and an identical fresh $2048 \to 128 \to 10$ classifier. Fine-tune for 20 epochs on either clean images or random crops (padding 4) and horizontal flips. Use Adam 0.001, reduced to 0.0001 at epoch 16. Freeze each CNN and standardize features using its original training inputs. For augmentation, cache two extra random views per image; each head-training batch chooses one of three views. These are repeated views, not independent samples. Both conditions have 100 root epochs, batch size 256, Adam 0.001 reduced at epoch 51, and equal optimizer-step counts. Evaluation uses unaugmented images.

**Correction tree.** Children share features and adjust root scores; they do not exclusively route samples. A raises a missed true-class score; B lowers an incorrectly winning class score. Three width-16 children are trained on the four remaining augmented-root errors with weighted BCE. Training-only checkpoint/strength selection retains bird A:

$$s'_c = s_c + a_c \mathbf{1}[g_{A,c} \geq 0] - b_c \mathbf{1}[g_{B,c} \geq 0], \qquad a_{\text{bird}} = 3.521489.$$

It fixes four training errors. Airplane B and deer B have zero strength. Clean roots already fit perfectly. CNN plus roots contain 2,629,098 parameters; adding the retained bird child and scalar gives 2,661,900. The CNN itself has 5,088 parameters.

**Implementation.** Jupyter, PyTorch 2.13.0, MPS on a MacBook Air; seed-42 full-data roots and ensemble seeds 101–105. Evidence is saved in the updated `TREE_AUGMENTATION_ENSEMBLE(1).ipynb`: current-assignment CIFAR experiments begin near cell 117 and Cats vs Dogs occupies cells 141–161 (zero-based). The supplied CSVs and final test JSON support the results. The copied notebook retains earlier work. Fresh execution requires the earlier CNN checkpoint and local dataset paths; it was inspected, not retrained for this report.

Public code repository: https://github.com/aromaindus049-byte/neural-tree-assignment.

## 2. Augmentation, interpolation and generalization

| Model (epoch 100) | Clean training accuracy | Validation | Test |
|---|---|---|---|
| Clean roots | 100.0000% | 71.88% | 72.37% |
| Augmented roots | 99.9911% | 74.38% | 74.45% |
| Augmented tree | 100.0000% | 74.38% | 74.45% |

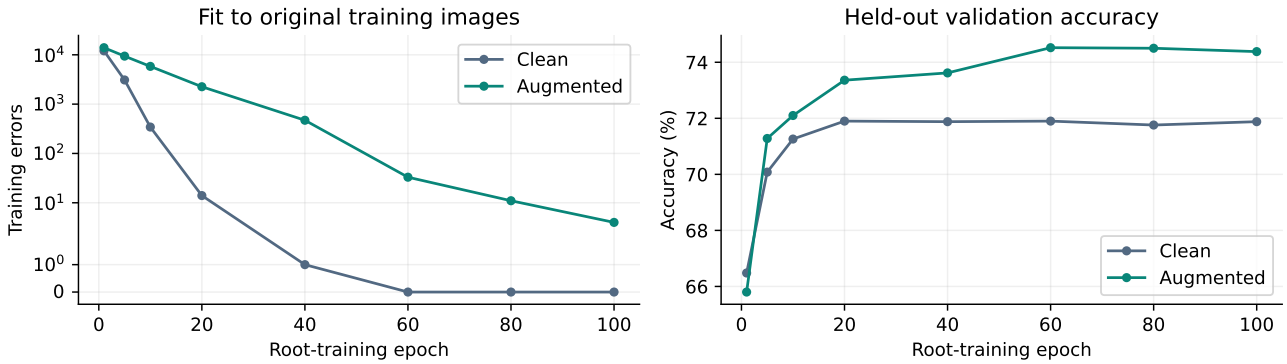



Figure 1. Recorded root-training curves on original inputs. The augmented correction child is added after epoch 100 and is not part of these root-only curves.

Augmentation improves test accuracy by **2.08 percentage points** (208 fewer errors), while the correction tree interpolates the original training set. Exact empirical fitting and improved generalization therefore coexist. This does not mean fitting every possible augmentation, nor that overfitting causes improvement. Augmentation affects both CNN and roots; their separate effects are not isolated.

**Double descent.** Clean roots first have zero measured errors at epoch 60; augmented roots retain four at epoch 100 and reach zero only after correction. Validation accuracy generally improves and plateaus. No convincing error peak near interpolation followed by a sustained second decline appears. A variance hump alone is not double descent in prediction risk. Sparse checkpoints and the epoch-51 learning-rate change limit interpretation. A subsequent three-seed capacity sweep with controlled label corruption is reported in Section 7; it also provides no convincing double-descent peak.

## 3. Ensemble behavior and majority voting

Each member uses 45,000 bootstrap draws (about 28,494 distinct training images), its own initialization/batch order, and the shared augmented CNN. Every member fits its own sampled clean inputs perfectly, so correction growth stops. Thus these members are **root-only forests from the correction-tree architecture**, not five nontrivial branched trees. Accuracy near 90% on the full training pool includes omitted bootstrap inputs.

| Members | Validation | Test | Test ties |
|---|---|---|---|
| 1 | 72.46% (mean of 5) | 71.93% (seed 101) | 0 |
| 3 | 73.97% (mean of 10 groups) | 74.25% (101–103) | 410 |
| 5 | 74.76% | 74.99% | 351 |

The class with the most votes wins (plurality voting, conventionally called majority voting here). Multiclass ties occur even for odd member counts; the smallest class index breaks ties. Mean pairwise validation disagreement is 20.93%. Five members gain 3.06 test points over seed 101 and 0.54 points (54 images) over the augmented full-data tree. No significance test was performed. Validation subgroups overlap and are not independent replications. Test uses predetermined prefix order and epoch 100, not a test-selected group.

## 4. Bias and variance across epochs and sample sizes

For $M = 5$ models and $N = 5,000$ validation inputs, let $h_{mi}$ be a one-hot class prediction, $y_i$ the observed one-hot label and $\bar{h}_i = M^{-1} \sum_m h_{mi}$. Define

$$B_{\text{proxy}} = \frac{1}{2N} \sum_i \|\bar{h}_i - y_i\|^2, \qquad V = \frac{1}{2NM} \sum_{i,m} \|h_{mi} - \bar{h}_i\|^2,$$

$$R_{\text{member}} = \frac{1}{NM} \sum_{i,m} \mathbf{1}[h_{mi} \neq y_i] \qquad = B_{\text{proxy}} + V.$$

Expand $h_{mi} - y_i = (h_{mi} - \bar{h}_i) + (\bar{h}_i - y_i)$. The cross term vanishes since $\sum_m (h_{mi} - \bar{h}_i) = 0$; distinct one-hot vectors have squared distance 2. The code checks the identity within $10^{-10}$ before printing rounded results. This is an exact finite-ensemble decomposition relative to observed labels, *not* estimated population bias or irreducible noise. Majority-vote error is separate.

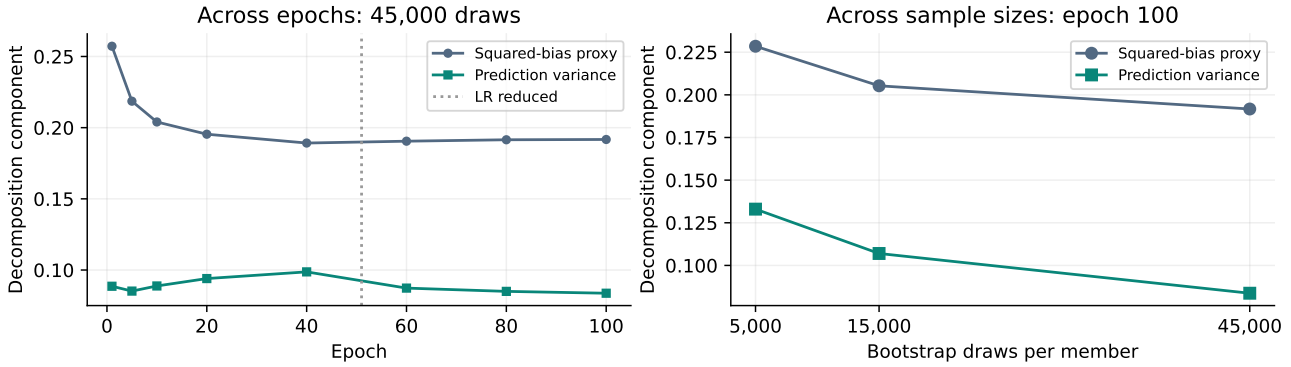



Figure 2. Components conditional on one shared CNN and cached views. Units are fractions. The vertical line marks the scheduled learning-rate reduction.

**Across epochs.** The proxy falls from 0.2572 (epoch 1) to 0.1892 (40), then reaches 0.1917 (100). Variance rises from 0.0886 to 0.0987, then falls to 0.0837. Mean member error falls from 34.59% to 27.54%. Voting peaks at 75.18% validation at epoch 60; the planned epoch-100 endpoint is retained for test. The variance decline coincides with the lower learning rate, without establishing causation.

| Draws/member | Mean distinct | $B_{\text{proxy}}$ | $V$ | Member val. | Vote val. |
|---|---|---|---|---|---|
| 5,000 | 4,739.2 | 0.2285 | 0.1330 | 63.84% | 68.48% |
| 15,000 | 12,800.8 | 0.2053 | 0.1070 | 68.78% | 72.48% |
| 45,000 | 28,493.6 | 0.1917 | 0.0837 | 72.46% | 74.76% |

All smaller-sample members also fit their distinct sampled clean images with zero errors at epoch 100. More head-training draws lower both components and improve validation. Voting ties fall from 377 to 295 to 179. Interpolation alone therefore does not determine generalization.

**Scope.** The CNN and normalization used all 45,000 training inputs (and labels during CNN training). This is a *head-training sample-size study*, not end-to-end training on only 5,000 or 15,000 images. Samples are nested prefixes of each seed's bootstrap draws; models share CNN errors. At equal epochs, sample sizes also receive different optimizer-step counts: 2,000/5,900/17,600 at epoch 100. Size and compute are not isolated. Five models provide empirical diagnostics, not complete population estimates.

## 5. Local loss landscape

Take the full-data augmented bird root before hard correction. Hold the CNN, normalization and biases fixed. Sample 2,048 original training and 2,048 validation images independently of predictions. Evaluate weighted bird-versus-rest BCE on a $17 \times 17$ grid:

$$W(\alpha, \beta) = W_0 + \alpha D_1 + \beta D_2, \qquad \alpha, \beta \in [-0.25, 0.25].$$

Two seeded Gaussian directions are normalized row-wise to trained weight norms; biases are not perturbed. This is a smooth-node loss slice, not a differentiable landscape for hard flags or majority votes.

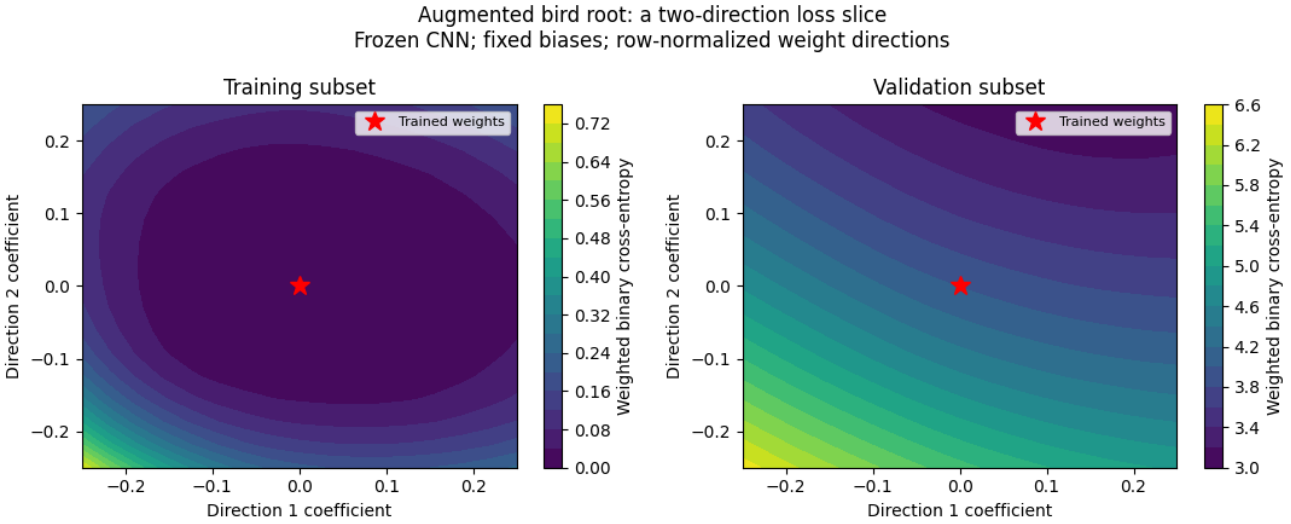



Figure 3. Red star: trained weights. Panels have separate color scales. Apparent basin width depends on the sampled directions, normalization and plotted range.

| Weighted BCE | Trained weights | Minimum sampled on grid |
|---|---|---|
| Training subset | 0.001523 | 0.001523 |
| Validation subset | 3.990316 | 3.052272 |

The trained point has the lowest sampled training loss; larger perturbations raise it. Validation loss prefers a displaced region, showing that empirical fitting need not optimize held-out loss. This does not establish global convexity, a full-space local minimum, Hessian sharpness or a causal flatness–accuracy relation. No new weights were selected from the plot.

## 6. CIFAR findings and extensions

**Practical contributions.** Cached multi-view features make augmentation-plus-interpolation experiments inexpensive; task-driven correction retains only a useful rare-error branch; bootstrap voting shares a CNN while adding diverse heads. An exact finite-ensemble identity separates changing systematic prediction error from variability. These are implemented extensions, not claims of a newly invented algorithm.

**Conclusions.** Augmentation improves held-out accuracy while the correction tree attains 100% training accuracy. Voting improves the observed test result to 74.99%. Both empirical decomposition components decrease with more head-training data; variance is nonmonotonic across epochs. Training and validation prefer different weights in the measured slice. Clear double descent is *not* observed.

**Limits and next steps.** One full-data head seed per condition. Cats vs Dogs extensions follow in Sections 8–10. The test set was already used in the earlier assignment, so it is not newly untouched; current models were fixed before evaluation. Ensemble members have no retained correction children. Next: deeper branched ensembles, trunks retrained within each sample-size study, augmentation-stage ablations, equal-compute comparisons, more seeds and denser capacity/epoch sampling near interpolation, and multiple landscape directions/nodes. A paired prediction analysis is needed before claiming a reliable advantage from 54 extra correct test decisions.

## 7. Extended double-descent investigation: three head seeds

**Protocol.** Fix a class-balanced subset of 5,000 original CIFAR training images (500/class), retaining all three cached feature views. Fit normalization on this subset's original features; the augmented CNN remains pretrained on the full training pool. Compare correct labels with exactly 750 labels (15%) changed to a different class using a fixed corruption mask. All views of an image receive the same assigned label. Validation labels remain clean. Vary root hidden width over $1, 2, 3, 4, 6, 8, 12, 16, 32, 64, 128, 512$. Ten roots have $20,500h + 10$ trainable parameters; the frozen CNN is excluded from this capacity axis.

Use seeds 201–203 on the same images, labels and views. Train each configuration for 300 epochs at Adam learning rate 0.001, then uniformly continue all configurations for 300 epochs at 0.0001, with batch size 256. Record metrics every 25 epochs. The continuation was introduced after inspecting seed 201's optimization instability, so this is a documented two-stage follow-up, not an initially fixed-rate convergence study. No validation-based checkpoint selection or additional test evaluation is used.

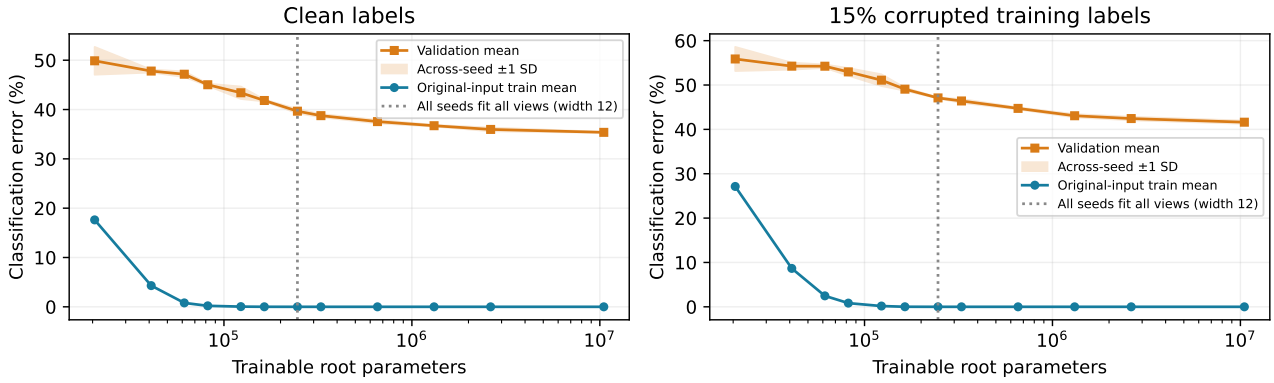



Figure 4. Final epoch-600 error across three head seeds. Shading is sample standard deviation, not a confidence interval. The dotted line marks the smallest tested width fitting all cached views for every seed. Tiny training errors can be hidden by the scale.

| At epoch 600 | Clean labels | 15% corrupted labels |
|---|---|---|
| Smallest final-zero original-input width (201/202/203) | 8/12/8 | 8/8/12 |
| Smallest final-zero all-view width (201/202/203) | 8/12/8 | 12/12/12 |
| Width 12 mean validation accuracy ± SD | $60.33 \pm 0.43\%$ | $52.91 \pm 0.10\%$ |
| Width 512 mean validation accuracy ± SD | $64.63 \pm 0.10\%$ | $58.38 \pm 0.35\%$ |

**Findings.** Mean validation error generally decreases with capacity, including beyond interpolation. No convincing peak near interpolation followed by a second decline is observed under either label condition. A few fluctuations at narrow widths occur before interpolation and are insufficient evidence for double descent. Wider models can fit corrupted labels yet generalize better than smaller interpolating models; label noise still lowers held-out accuracy.

**Optimization matters.** In seed 201, clean widths 8 and 12 lost fit late in the first stage; lower-rate continuation restored it. Width 8 ultimately fits all views for two clean seeds, but seed 202 retains one error. In the noisy case width 8 has 0/0/2 original-input errors and 1/1/3 all-view errors; width 12 fits every view for all seeds. Thus "ever interpolated," "final original-input interpolation" and "final all-view interpolation" are distinct. The smallest tested fitting width is protocol-dependent, not a theoretical threshold.

**Limits.** Narrow models still show falling loss and are not proven converged. Three seeds vary head initialization and batch/view order only; the subset, corruption realization and CNN are fixed. This is a conditional neural-head capacity study, not a new tree-depth sweep or an end-to-end 5,000-image experiment. The 72-row CSV verifies the final measurements; the updated notebook contains the sweep and should accompany the submission. This extension strengthens a negative finding, rather than demonstrating universal absence of double descent.

## 8. Cats vs Dogs: augmentation, contradiction-aware trees and voting

**Data and architecture.** Use the previously audited RGB $32 \times 32$ split: 19,998 training images (9,999/class), 2,494 validation images (1,246 cats; 1,248 dogs), and 2,496 test images (1,245 cats; 1,251 dogs). The earlier audit excluded four validation and six test copies matching inputs in other splits. No identical audited preprocessed inputs cross these revised splits. This does not rule out near-duplicates. Two conflicting-label groups remain in training: identical preprocessed inputs carry opposite labels, forcing at least two errors for any deterministic classifier of these inputs.

The shared CNN has the same 5,088-parameter two-block architecture. A single binary root is $2048 \to 128 \to 1$ (262,401 parameters), with Cat=0 and Dog=1, sign threshold 0 and unweighted BCE. Clean and augmented conditions fine-tune the earlier Cats vs Dogs CNN for 20 epochs with identical fresh heads, then freeze it. Crops, flips, two cached augmented views, standardization, root epochs and learning-rate schedules match the CIFAR protocol. They are matched continuations from an earlier trained CNN, not independent training from scratch.

| Epoch-100 model | Training | Validation | Test | Test errors |
|---|---|---|---|---|
| Clean full-data root | 99.9900% | 80.83% | 80.05% | 498 |
| Augmented full-data root | 99.9900% | 81.80% | 81.97% | 450 |
| One bootstrap tree (101) | – | 81.40% | 81.57% | 460 |
| Three-tree vote (101–103) | – | – | 82.09% | 447 |
| Five-tree vote (101–105) | – | 83.04% | 82.85% | 428 |

Test balanced accuracies round to the same two-decimal values. Dashes distinguish unreported combinations from full-data training measurements.

Augmentation removes 48 test errors, a 1.92-point gain. Five-tree voting removes another 22 errors versus the full-data augmented root, a 0.88-point gain; it improves on the clean baseline by 2.80 points. These are descriptive differences, without a paired significance analysis. The predefined seed order and epoch 100 are retained for evaluation; capacity experiments do not select test models. This split derives from the dataset used in the earlier assignment, rather than a newly collected independent test set.

**Contradiction-aware branch growth.** Bootstrap each member with 19,998 draws (mean 12,631 distinct images). Minimum weighted sampled errors are computed by grouping identical features and summing minority-label multiplicities. Seeds 101–105 have root errors $1, 0, 4, 0, 5$ and lower bounds $1, 0, 2, 0, 2$. Grow only branches correcting errors above these bounds. A width-16 B child for seed 103 and A child for seed 105 reduce errors to $1, 0, 2, 0, 2$, respectively. Strengths 4.518374 and 0.233285 are selected on training data only. Thus all five trees attain their sampled contradiction floors; three remain root-only. Each retained child has 32,801 parameters plus its strength scalar.

No validation predictions change after correction. Root and corrected-tree test accuracies and error counts are equal for each reported vote size; equal aggregate metrics alone do not prove identical individual test predictions. The branches demonstrate targeted memorization, while observed held-out gains come from augmentation and voting. Binary voting with 1, 3 or 5 members has no ties. Across all validation combinations, mean accuracy rises from 81.05% (one) to 82.25% (three), then 83.04% (five); mean pairwise disagreement is 11.64%.

## 9. Cats vs Dogs: bias–variance diagnostics and loss slice

Apply the same one-hot finite-ensemble decomposition in Section 4 with two classes and $N = 2,494$. These diagnostics use the bootstrap roots before correction. More head-training draws consistently reduce empirical prediction variance; the squared-bias proxy decreases initially but is not strictly monotonic.

| Draws/member | Mean distinct | $B_{\text{proxy}}$ | $V$ | Member val. | Vote val. |
|---|---|---|---|---|---|
| 2,000 | 1,899.4 | 0.1472 | 0.0620 | 79.08% | 80.83% |
| 6,000 | 5,183.4 | 0.1418 | 0.0549 | 80.33% | 82.32% |
| 19,998 | 12,631.0 | 0.1429 | 0.0466 | 81.05% | 83.04% |

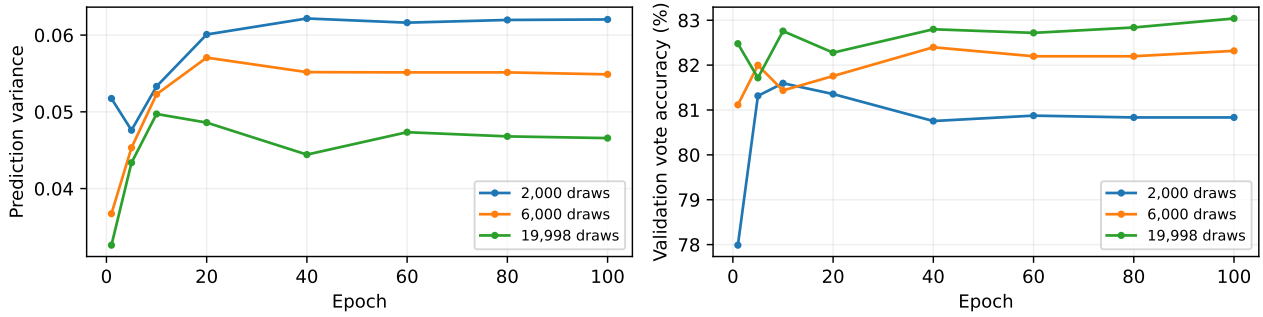



Figure 5. Five-head empirical diagnostics conditional on the shared augmented CNN. Different sample sizes receive different optimizer-step counts at equal epochs.

For full-size draws, the proxy changes from 0.1546 at epoch 1 to 0.1395 at epoch 10, ending at 0.1429. Variance rises from 0.0326 to 0.0497 at epoch 10 and ends at 0.0466. Vote accuracy varies nonmonotonically, from 82.48% to 83.04%. Neither bias nor variance must decrease every epoch. Smaller-sample members fit their distinct clean sampled inputs perfectly. The CNN and normalization nevertheless use the full training pool; this measures conditional head behavior, not end-to-end learning from only 2,000 images.

**Local loss landscape.** Perturb the augmented full-data binary root using two seeded, row-normalized Gaussian weight directions. Fix the CNN, normalization and biases. Evaluate unweighted BCE on 2,048 original training and 2,048 validation inputs on a $17 \times 17$ grid over $[-0.25, 0.25]^2$. Restore original weights afterward.

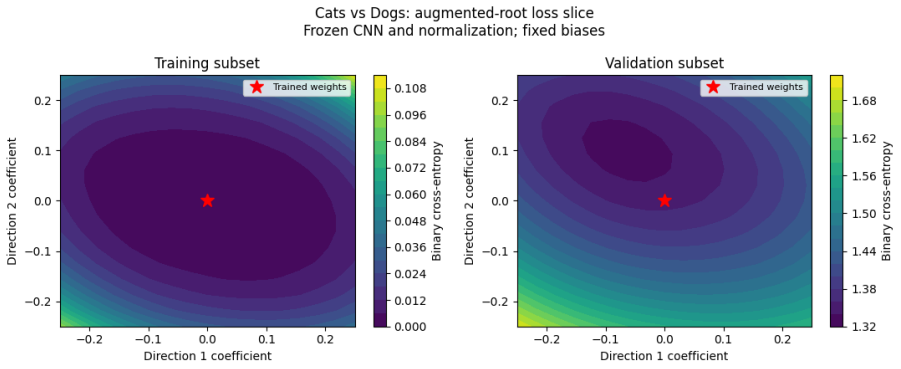



Figure 6. Local binary-root slice, with separate color scales. The red star marks the trained solution; sampled directions do not characterize the entire parameter space.

| Unweighted BCE | Trained weights | Minimum sampled |
|---|---|---|
| Training subset | 0.000808 | 0.000791 |
| Validation subset | 1.351692 | 1.332186 |

The trained point lies in a low-training-loss region but is not the exact grid minimum. Validation prefers displaced weights. This supports a training/validation objective mismatch; it does not prove global convexity or that a flat minimum causes generalization. Unlike CIFAR's weighted class-specific BCE, these loss values use unweighted binary BCE and should not be compared numerically across datasets.

## 10. Cats vs Dogs capacity study and combined conclusions

**Controlled follow-up.** Fix 5,000 training images (2,500/class) and three associated cached views. Compare original labels with exactly 750 binary labels flipped (15%); keep validation labels unchanged. Fit normalization on the selected original features. The augmented CNN was trained on the full training pool, so the experiment is conditional on a representation learned from more than 5,000 labels. Use seed 201 and widths $1, 2, 3, 4, 6, 8, 12, 16, 32, 64, 128$; a binary root contains $2050h + 1$ parameters, excluding the CNN. Each run uses 300 epochs at Adam 0.001, followed by 300 at 0.0001 after first-stage instability was inspected. No claim of full convergence is made.

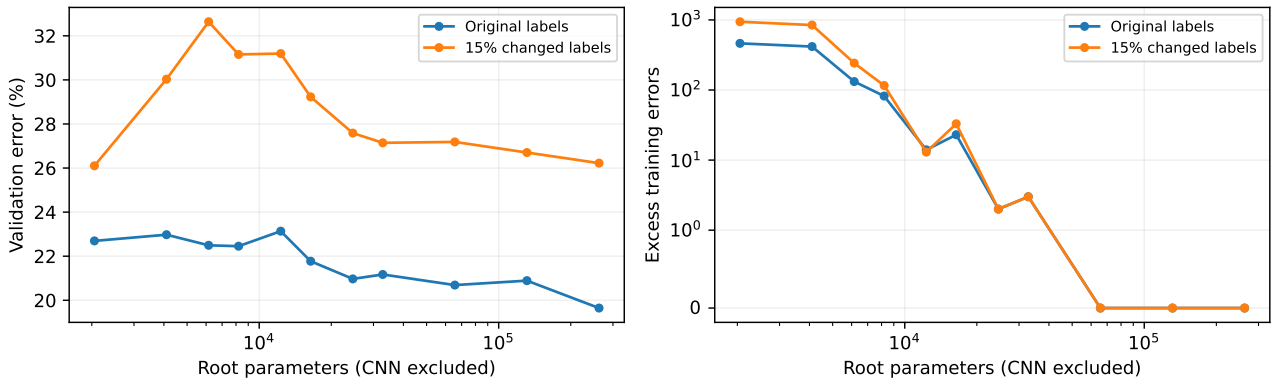



Figure 7. Final epoch-600 capacity curves from 22 CSV rows. The training ordinate subtracts the unavoidable one-error original-feature floor; reaching zero excess is not exact interpolation.

**What the curves establish.** Both label conditions have an original-feature contradiction floor of one error. At epoch 600, widths 32, 64 and 128 reach it and make one error across all three cached views. With 15% corrupted labels, validation accuracy is 73.90% at width 1, falls to 67.36% at width 3, then recovers to 73.78% at width 128. However, the worst point occurs well before the smallest tested floor-reaching width (32). This is a nonmonotonic capacity curve, *not confirmed interpolation-centered double descent*. The initial descent needed for a full double-descent curve is not established by these tested widths. Clean-label accuracy is 77.31% at width 1 and 80.35% at width 128. Only one head seed is available for this dataset.

**Implemented innovations and interpretations.**

- Shared cached original/augmented views make repeated fitting and ensemble experiments practical on an MPS laptop. Augmentation and near-perfect empirical fitting coexist with improved held-out accuracy on both datasets.
- Contradiction-aware stopping avoids futile tree growth against identical inputs with opposite labels. It separates irreducible empirical conflicts from correctable model errors; it is not an estimate of population Bayes noise.
- Training-driven branches fit rare residual errors, but show little or no held-out benefit. Diverse-head voting gives the larger observed gain, reaching 74.99% CIFAR and 82.85% Cats vs Dogs test accuracy.
- Larger head-training samples reduce conditional prediction variance on both datasets. Bias proxies need not decrease monotonically, and epoch-dependent variance humps do not by themselves establish double descent.

**Limitations and future directions.** Repeat Cats vs Dogs capacity sweeps across more seeds, subsets and corruption masks; sample more densely around the floor-reaching region and separate optimization failure from insufficient capacity. Test whether conflict-free, explicitly documented subsets permit a cleaner interpolation study. Retrain trunks within each sample-size condition, compare equal optimizer-step budgets, and ablate augmentation at the CNN and node stages separately. Explore independently trained trunks for ensemble diversity, deeper correction trees and their size/accuracy tradeoffs. Repeat loss slices across directions and nodes, and evaluate probability calibration alongside accuracy. Use paired prediction statistics before asserting reliable ensemble gains. No current evidence warrants claiming universal double descent or that memorizing labels improves generalization by itself.

Reproducibility: the updated notebook contains code and saved outputs; it has been inspected, not executed end to end here. Earlier checkpoints and dataset manifests remain dependencies. Keep CIFAR and Cats vs Dogs result files in separate folders because several basenames are identical.